

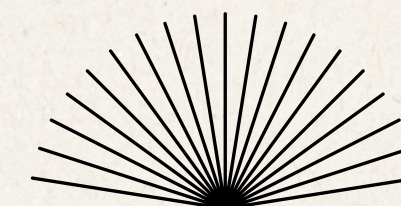


PROJECT 2 | SECP3623

CAR RENTAL SYSTEM

Group 1

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Introduction

- Manages **vehicles, customers, booking and payments**
- Customer can:
 - Rent vehicles for specific periods
 - Track booking
 - Make payments (deposits & refunds)

Project Objectives

- Design **NoSQL backend** using MongoDB
- Apply document data modelling principles:
 - Embedding
 - Referencing
 - Arrays
- Demonstrate **CRUD operations**
- Implement **indexing** and **aggregation pipelines**
- Analyse **security considerations** and **design challenges**

Aspect	Embedding	Referencing
Purpose	For tightly related data	For shared data across collections
Data Access	Faster reads (single document query)	Requires joins / \$lookup
Data Duplication	Acceptable and common	Avoided
Relationship Type	Strongly dependent	Loosely coupled
Example in System	pickup, return, deposit, refund	booking → customer, booking → vehicle, payment → booking

CRUD Implementation

- Insert Operation (create)
 - **insertOne()**
 - **insertMany()**
- Query Operation (read)
 - find()
 - findOne()
 - Conditional Filtering
 - **\$gt**
 - **\$gte**
 - **\$in**
 - **\$and**
 - **\$or**
 - **include/exclude field**
- Update Operation
 - **updateOne()**
 - updateMany()
 - **\$inc**
 - \$push
 - **\$pull**
- Delete Operation
 - deleteOne()
 - deleteMany()

Indexing Implementation

- Created indexes on frequently queried fields in the vehicles collection
- Single-field index on vehicle_status
- Compound index on vehicle_location and vehicle_status

Purpose of Indexing

- Improve data retrieval speed
- Reduce full collection scans
- Support faster vehicle availability searches

Query Performance Impact

- Indexed queries return results more efficiently
 - Improved scalability as data volume increases
 - Enhances overall system responsiveness
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Aggregation Pipeline

Total revenue for “paid” bookings grouped by payment method

- \$match
- \$group

Total payment amount within a time range, grouped by payment status

- \$match
- \$group

Bookings with non paid payment statuses

- \$match
- \$lookup
- \$unwind
- project

Deposit refund status count and total deposit amount

- \$group

Customers with forfeited deposits

- \$match
- \$lookup
- \$unwind
- project

Most rented vehicle class

- \$match
- \$lookup
- \$unwind
- \$group
- \$sort

The top 3 vehicle make owned

- \$group
- \$sort
- \$limit

Sorting demonstrations

Purpose of Sorting

- Improve data readability
- Support efficient data analysis
- Enhance system usability and reporting

Example

- Sorting Customers' Contact Information by Name (Ascending)
 - Sorting Paid Payment by Amount (Descending)
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Performance Improvements

by Advanced Operations

Indexing

- Reduces full collection scans
- Enables faster data retrieval on frequently queried fields

Aggregation Pipelines

- Processes and summarizes data within MongoDB
- Reduces unnecessary data transfer and improves analytics efficiency

Sorting

- Organizes query results in meaningful order
 - Improves usability after filtering operations
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Team Member Contribution

Team Member	Role and Responsibilities
Teh Ru Qian	• Provide the project overview, domain description, objectives • Design the NoSQL data model
Tan Yi Ya	• Implement the basic MongoDB CRUD operations • Summarize the project outcomes
Woo Cheng Shuan	• Design and demonstrate indexing strategies • Analyze query performance and optimizing retrieval in MongoDB
Lam Yoke Yu	• Create and demonstrate aggregation pipelines to perform complex analytics • Generate operational insights from the database
Goe Jie Ying	• Implement sorting operations • Discuss security measures for sensitive data and analyze the limitations of MongoDB

**THANK YOU
FOR LISTENING!**
